

Assignment 3

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1 Set up

Consider the following damped oscillation.

$$m \frac{dv}{dt} = -\zeta v - kx \quad (1)$$

$$v = \frac{dx}{dt} \quad (2)$$

by substitute (2) in to (1) we get second order ODE.

$$m \frac{d^2}{dt^2} x + \zeta \frac{dx}{dt} + kx = 0 \quad (3)$$

which has characteristic equation

$$m\lambda^2 + \zeta\lambda + k = 0 \quad (4)$$

We can characterise behavior of the equation by different λ ; Namely,

$$\begin{cases} \zeta^2 - 4km > 0 & \text{overdamped} \\ \zeta^2 - 4km < 0 & \text{underdamped} \\ \zeta^2 - 4km = 0 & \text{critically damped} \end{cases} \quad (5)$$

Let's dedimensionalize (1) by the following

$$\begin{aligned} x &= a\tilde{x} \\ v &= \frac{a}{t_0}\tilde{v} \\ t &= t_0\tilde{t} \end{aligned} \quad (6)$$

where $\tilde{x}, \tilde{v}, \tilde{t}$ are dimensionless.

$$\begin{aligned} m \frac{a}{t_0} \frac{d\tilde{v}}{t_0 d\tilde{t}} &= -\zeta \frac{a}{t_0} \tilde{v} - ka\tilde{x} \\ 0 &= \frac{d\tilde{v}}{d\tilde{t}} + \frac{\zeta}{m} t_0 \tilde{v} + \frac{k}{m} t_0^2 \tilde{x} \end{aligned}$$

By substitute $t_d = m/\zeta$ and $t_s^2 = m/k$ we get

$$0 = \frac{d\tilde{v}}{d\tilde{t}} + \frac{t_0}{t_d}\tilde{v} + \frac{t_0^2}{t_s^2}\tilde{x}$$

Especially when choosing $t_0 = t_s$ and define $\tau = t_s/t_d$ we get

$$0 = \frac{d}{d\tilde{t}}\tilde{v} + \tau\tilde{v} + \tilde{x} \tag{7}$$

$$\tilde{v} = \frac{d}{d\tilde{t}}\tilde{x} \tag{8}$$

with parameter $\tau > 0$. Hence, we rewrite all possible outcomes in term of dimensionless parameter.

$$\begin{cases} \tau > 2 & \text{overdamped} \\ \tau < 2 & \text{underdamped} \\ \tau = 2 & \text{critically damped} \end{cases} \tag{9}$$

2 Result

With the following initial conditions

$$\begin{aligned} x(0) &= a \\ v(0) &= 0 \end{aligned} \tag{10}$$

or in dimensionless parameter

$$\begin{aligned} \tilde{x}(0) &= 1 \\ \tilde{v}(0) &= 0 \end{aligned} \tag{11}$$

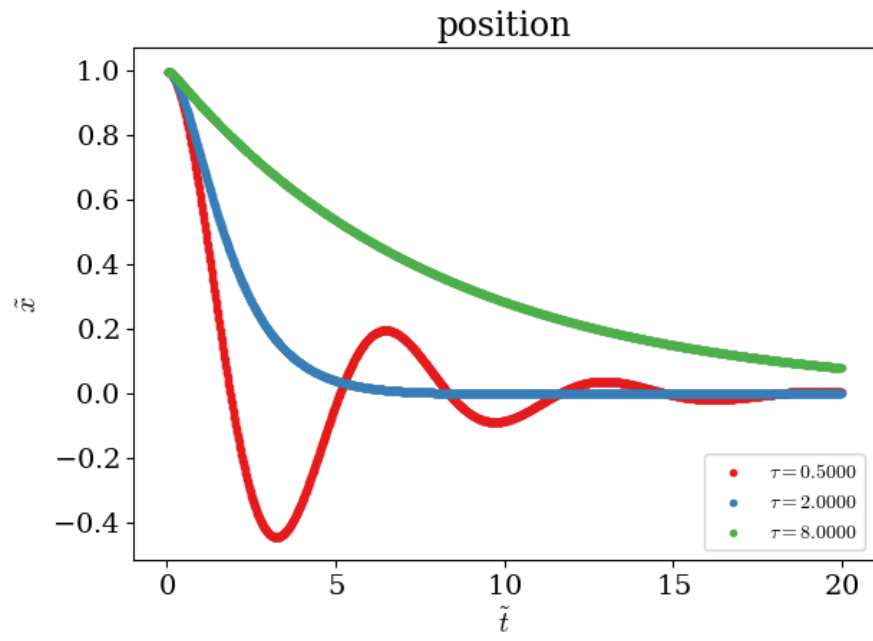


Figure 1: Result

Clearly, the behavior of oscillation is dictated by dimensionless parameter τ .